



南京航空航天大学  
NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS

## 《大模型安全与知识增强》实验报告

题 目： 知识编辑实验

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## 1. 实验目的

本实验的主要目的是学习和实践大模型知识编辑方法。随着大语言模型在问答、检索和智能助手等场景中的应用越来越广泛，模型中可能会存在知识过期或者事实错误的问题。如果每次都通过重新训练来修正这些问题，成本会很高，因此知识编辑提供了一种更高效的思路，即在尽量不影响模型整体能力的前提下，修改模型中的特定事实知识。

本实验基于 Qwen2.5-0.5B-Instruct 模型和 EasyEdit 框架，主要完成以下内容：对原始模型进行基线测试；使用 ROME 方法进行单条知识编辑；使用 MEMIT 方法进行批量知识编辑；对实验结果进行综合评估和分析。

## 2. 实验环境与数据集

### 2.1 实验环境

操作系统：Windows

Python 环境：easyedit

基础模型：Qwen2.5-0.5B-Instruct

主要工具：PyTorch、Transformers、EasyEdit

### 2.2 数据集构建

本实验根据要求手动构建了一个包含 10 条事实更新样本的数据集，保存在 data/custom\_edits.json 中。每条样本包括以下字段：prompt、target\_new、ground\_truth、rephrase\_prompt、locality\_prompt、locality\_ground\_truth。

这些样本主要涉及公司 CEO、国家首都、国家元首以及球队主教练等事实信息，用于测试模型在知识编辑前后的变化情况。

## 3. Task 1：基线测试

在不对模型进行任何编辑的情况下，首先使用 baseline.py 对原始模型进行测试，判断模型是否已经掌握目标新知识，以及是否仍然保留旧知识。

### 3.1 实验结果

```
(easyedit) E:\course\LLM\03>python baseline.py
E:\ToolsData\AnacondaEnv\envs\easyedit\lib\site-packages\transformers\utils\hub.py:110: FutureWarning: Using 'TRANSFORMERS_CACHE' is deprecated and will be removed in v5 of Transformers. Use 'HF_HOME' instead.
  warnings.warn(
Loading model: E:\course\LLM\03\model\Qwen2.5-0.5B-Instruct

[1/10] Prompt: The CEO of Starbucks is
The following generation flags are not valid and may be ignored: ['temperature', 'top.p', 'top.k']. Set 'TRANSFORMERS_VERBOSITY=info' for more details.
Model answer: a man named Howard Schultz. He was born in 1964 and he has been the CEO of Starbucks since 2003.

Now

[2/10] Prompt: The CEO of Intel is
Model answer: a man named Pat Gelsinger. He has been the CEO of Intel since 2013. In his first year, he led the company to

[3/10] Prompt: The monarch of the United Kingdom is
Model answer: Queen Elizabeth II. She was born on 21 January 1926 in Windsor Castle, Berkshire, England. Her father, King George VI,

[4/10] Prompt: The CEO of Twitter is
Model answer: a man named Jack Dorsey. He was born on January 1, 1972 in New York City, and he has been the CEO of

[5/10] Prompt: The head coach of FC Barcelona is
Model answer: a man named Luis Enrique. He has been the head coach of the club since 2013 and he was born on January 1, 1

[6/10] Prompt: The CEO of YouTube is
Model answer: planning to introduce a new feature that will allow users to upload videos in different languages. The company has decided to use the Google Translate API for this purpose, which

[7/10] Prompt: The CEO of IBM is
Model answer: planning to distribute a total of $10,000 among three employees: John, Jane, and Tom. The distribution must follow these rules:
1

[8/10] Prompt: The capital of Turkey is
Model answer: located in the northwestern part of the country. The city has a population of 1,057,426 people and an area of

[9/10] Prompt: The CEO of Google is
Model answer: a man named Larry Page. He was born on January 1, 1956 in Mountain View, California. His father, Sergey, was also

[10/10] Prompt: The CEO of AMD is
Model answer: a man named John. He has been with the company for 20 years and has held various positions within the company, including Chief Technology Officer (CTO)

==== Baseline Summary ====
Total samples: 10
Already matches target_new: 0/10
Matches ground_truth: 6/10
Saved to: results\baseline_results.json
```

总样本数：10

已匹配 target\_new：0/10

匹配 ground\_truth：6/10

## 3.2 结果分析

实验结果说明，原始模型基本不具备目标新知识，同时仍保留一定比例的旧知识。因此，这些样本适合作为知识编辑实验的测试集，也说明后续的知识编辑任务是有必要的。

## 4. Task 2：ROME 单条知识编辑

在这一部分，使用 ROME 方法对 10 条事实逐条进行单条知识编辑。由于 ROME 更适合单条知识修改，因此每次只编辑一条样本，并在编辑后重置模型权重，避免不同样本之间相互影响。

### 4.1 评估结果

|                                                                                                                                                                                                                                 |                |     |        |     |    |         |     |     |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|--------|-----|----|---------|-----|-----|-----|
| Table 1: Summary of the data for the first 1000 rows. The table contains 10 columns: ID, Name, Age, Gender, Height, Weight, Blood Pressure, Heart Rate, Cholesterol, and Glucose. The data is sorted by Age in ascending order. |                |     |        |     |    |         |     |     |     |
| 1                                                                                                                                                                                                                               | John Doe       | 25  | Male   | 175 | 70 | 120/80  | 75  | 180 | 100 |
| 2                                                                                                                                                                                                                               | Jane Smith     | 30  | Female | 160 | 55 | 110/70  | 65  | 160 | 90  |
| 3                                                                                                                                                                                                                               | Bob Johnson    | 45  | Male   | 180 | 85 | 130/90  | 85  | 200 | 110 |
| 4                                                                                                                                                                                                                               | Alice Brown    | 50  | Female | 165 | 60 | 125/85  | 70  | 170 | 95  |
| 5                                                                                                                                                                                                                               | Charlie Wilson | 55  | Male   | 170 | 75 | 135/95  | 90  | 210 | 120 |
| 6                                                                                                                                                                                                                               | Diana Prince   | 60  | Female | 155 | 50 | 115/75  | 60  | 150 | 85  |
| 7                                                                                                                                                                                                                               | Frank Miller   | 65  | Male   | 175 | 80 | 140/100 | 100 | 220 | 130 |
| 8                                                                                                                                                                                                                               | Grace Lee      | 70  | Female | 160 | 65 | 120/80  | 75  | 180 | 100 |
| 9                                                                                                                                                                                                                               | Henry Davis    | 75  | Male   | 170 | 70 | 130/90  | 85  | 190 | 110 |
| 10                                                                                                                                                                                                                              | Ivy Clark      | 80  | Female | 150 | 45 | 110/70  | 60  | 140 | 80  |
| 11                                                                                                                                                                                                                              | Jack Black     | 85  | Male   | 165 | 55 | 125/85  | 70  | 170 | 95  |
| 12                                                                                                                                                                                                                              | Karen White    | 90  | Female | 155 | 50 | 115/75  | 60  | 150 | 85  |
| 13                                                                                                                                                                                                                              | Liam Green     | 95  | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 14                                                                                                                                                                                                                              | Mia Grey       | 100 | Female | 160 | 55 | 110/70  | 65  | 150 | 85  |
| 15                                                                                                                                                                                                                              | Noah Blue      | 105 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 16                                                                                                                                                                                                                              | Olivia Red     | 110 | Female | 165 | 60 | 120/80  | 70  | 160 | 90  |
| 17                                                                                                                                                                                                                              | Peter Yellow   | 115 | Male   | 170 | 65 | 125/85  | 75  | 170 | 95  |
| 18                                                                                                                                                                                                                              | Quinn Purple   | 120 | Female | 160 | 55 | 110/70  | 65  | 150 | 85  |
| 19                                                                                                                                                                                                                              | Ryan Silver    | 125 | Male   | 175 | 60 | 120/80  | 70  | 160 | 90  |
| 20                                                                                                                                                                                                                              | Sarah Gold     | 130 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 21                                                                                                                                                                                                                              | Tom Bronze     | 135 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 22                                                                                                                                                                                                                              | Uma Copper     | 140 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 23                                                                                                                                                                                                                              | Victor Iron    | 145 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 24                                                                                                                                                                                                                              | Wendy Nickel   | 150 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 25                                                                                                                                                                                                                              | Xavier Zinc    | 155 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 26                                                                                                                                                                                                                              | Yara Cadmium   | 160 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 27                                                                                                                                                                                                                              | Zoe Silver     | 165 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 28                                                                                                                                                                                                                              | Adam Gold      | 170 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 29                                                                                                                                                                                                                              | Eve Copper     | 175 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 30                                                                                                                                                                                                                              | Frank Silver   | 180 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 31                                                                                                                                                                                                                              | Grace Nickel   | 185 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 32                                                                                                                                                                                                                              | Henry Zinc     | 190 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 33                                                                                                                                                                                                                              | Ivy Cadmium    | 195 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 34                                                                                                                                                                                                                              | Jack Silver    | 200 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 35                                                                                                                                                                                                                              | Karen Gold     | 205 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 36                                                                                                                                                                                                                              | Liam Copper    | 210 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 37                                                                                                                                                                                                                              | Mia Silver     | 215 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 38                                                                                                                                                                                                                              | Noah Nickel    | 220 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 39                                                                                                                                                                                                                              | Olivia Zinc    | 225 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 40                                                                                                                                                                                                                              | Peter Cadmium  | 230 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 41                                                                                                                                                                                                                              | Quinn Silver   | 235 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 42                                                                                                                                                                                                                              | Ryan Gold      | 240 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 43                                                                                                                                                                                                                              | Sarah Copper   | 245 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 44                                                                                                                                                                                                                              | Tom Silver     | 250 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 45                                                                                                                                                                                                                              | Uma Nickel     | 255 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 46                                                                                                                                                                                                                              | Victor Zinc    | 260 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 47                                                                                                                                                                                                                              | Wendy Cadmium  | 265 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 48                                                                                                                                                                                                                              | Xavier Silver  | 270 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 49                                                                                                                                                                                                                              | Yara Gold      | 275 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 50                                                                                                                                                                                                                              | Zoe Copper     | 280 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 51                                                                                                                                                                                                                              | Adam Silver    | 285 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 52                                                                                                                                                                                                                              | Eve Nickel     | 290 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 53                                                                                                                                                                                                                              | Frank Zinc     | 295 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 54                                                                                                                                                                                                                              | Grace Cadmium  | 300 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 55                                                                                                                                                                                                                              | Henry Silver   | 305 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 56                                                                                                                                                                                                                              | Ivy Gold       | 310 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 57                                                                                                                                                                                                                              | Jack Copper    | 315 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 58                                                                                                                                                                                                                              | Karen Silver   | 320 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 59                                                                                                                                                                                                                              | Liam Nickel    | 325 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 60                                                                                                                                                                                                                              | Mia Zinc       | 330 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 61                                                                                                                                                                                                                              | Noah Cadmium   | 335 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 62                                                                                                                                                                                                                              | Olivia Silver  | 340 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 63                                                                                                                                                                                                                              | Peter Gold     | 345 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 64                                                                                                                                                                                                                              | Quinn Copper   | 350 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 65                                                                                                                                                                                                                              | Ryan Silver    | 355 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 66                                                                                                                                                                                                                              | Sarah Nickel   | 360 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 67                                                                                                                                                                                                                              | Tom Zinc       | 365 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 68                                                                                                                                                                                                                              | Uma Cadmium    | 370 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 69                                                                                                                                                                                                                              | Victor Silver  | 375 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 70                                                                                                                                                                                                                              | Wendy Gold     | 380 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 71                                                                                                                                                                                                                              | Xavier Copper  | 385 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 72                                                                                                                                                                                                                              | Yara Silver    | 390 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 73                                                                                                                                                                                                                              | Zoe Nickel     | 395 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 74                                                                                                                                                                                                                              | Adam Zinc      | 400 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 75                                                                                                                                                                                                                              | Eve Cadmium    | 405 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 76                                                                                                                                                                                                                              | Frank Silver   | 410 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 77                                                                                                                                                                                                                              | Grace Gold     | 415 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 78                                                                                                                                                                                                                              | Henry Copper   | 420 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 79                                                                                                                                                                                                                              | Ivy Silver     | 425 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 80                                                                                                                                                                                                                              | Jack Nickel    | 430 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 81                                                                                                                                                                                                                              | Karen Zinc     | 435 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 82                                                                                                                                                                                                                              | Liam Cadmium   | 440 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 83                                                                                                                                                                                                                              | Mia Silver     | 445 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 84                                                                                                                                                                                                                              | Noah Gold      | 450 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 85                                                                                                                                                                                                                              | Olivia Copper  | 455 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 86                                                                                                                                                                                                                              | Peter Silver   | 460 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 87                                                                                                                                                                                                                              | Quinn Nickel   | 465 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 88                                                                                                                                                                                                                              | Ryan Zinc      | 470 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 89                                                                                                                                                                                                                              | Sarah Cadmium  | 475 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 90                                                                                                                                                                                                                              | Tom Silver     | 480 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 91                                                                                                                                                                                                                              | Uma Gold       | 485 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 92                                                                                                                                                                                                                              | Victor Copper  | 490 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 93                                                                                                                                                                                                                              | Wendy Silver   | 495 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 94                                                                                                                                                                                                                              | Xavier Nickel  | 500 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 95                                                                                                                                                                                                                              | Yara Zinc      | 505 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 96                                                                                                                                                                                                                              | Zoe Cadmium    | 510 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |
| 97                                                                                                                                                                                                                              | Adam Silver    | 515 | Male   | 170 | 60 | 120/80  | 70  | 160 | 90  |
| 98                                                                                                                                                                                                                              | Eve Gold       | 520 | Female | 160 | 50 | 110/70  | 60  | 140 | 80  |
| 99                                                                                                                                                                                                                              | Frank Copper   | 525 | Male   | 175 | 65 | 125/85  | 75  | 170 | 95  |
| 100                                                                                                                                                                                                                             | Grace Silver   | 530 | Female | 165 | 55 | 115/75  | 65  | 155 | 88  |

ES/Efficacy: 100.00%  
PS/Generalization: 76.57%  
NS/Locality: 48.33%

## 4.2 结果分析

从结果可以看出，ROME 成功将 10 条目标知识全部写入模型，编辑成功率达到 100%。与此同时，模型在改写问题上的表现也较好，说明新知识在一定程度上实现了泛化，并不只是对原始 prompt 机械记忆。

不过，ROME 的局部性得分相对较低，说明在编辑目标知识的同时，也对部分无关事实造成了影响。这表明知识编辑虽然能够成功修改目标输出，但仍可能对模型内部其他知识造成扰动。

## 5. Task 3: MEMIT 批量知识编辑

在这一部分，使用 MEMIT 方法一次性对 500 条事实进行批量编辑。与 ROME 相比，MEMIT 更适合多条知识同时写入模型。

### 5.1 运行情况

由于 MEMIT 首次运行需要下载并处理 Wikipedia 数据集，同时为多个目标层计算 covariance statistics，因此整体耗时较长。本实验中，MEMIT 首次完整运行耗时为 2083.76 秒，峰值显存约为 3603.19MB。

### 5.2 评估结果

```

05/23/2026 16:47:28 - INFO - easyeditor.editors.editor - Evaluation took 4.578851938247681

===== Per-case MEMIT Metrics =====
[1/500] Vinson Massif is located in the continent of -> Antarctica
rewrite_acc: [np.float64(1.0)]
rephrase_acc: [np.float64(0.0)]
locality_acc: [np.float64(1.0)]
[2/500] Beats Music is owned by -> Apple
rewrite_acc: [np.float64(1.0)]
rephrase_acc: [np.float64(0.0)]
locality_acc: [np.float64(1.0)]
[3/500] Audible.com is owned by -> Amazon
rewrite_acc: [np.float64(1.0)]
rephrase_acc: [np.float64(0.0)]
locality_acc: [np.float64(1.0)]
[4/500] The Big Bang Theory premieres on -> CBS
rewrite_acc: [np.float64(1.0)]
rephrase_acc: [np.float64(1.0)]
locality_acc: [np.float64(1.0)]
[5/500] MacApp, a product created by -> Apple
rewrite_acc: [np.float64(1.0)]
rephrase_acc: [np.float64(1.0)]
locality_acc: [np.float64(1.0)]
[6/500] Giuseppe Angeli, who has a citizenship of -> Italy
rewrite_acc: [np.float64(1.0)]
rephrase_acc: [np.float64(1.0)]
locality_acc: [np.float64(1.0)]
[7/500] Catalonia belongs to the continent of -> Europe
rewrite_acc: [np.float64(1.0)]
rephrase_acc: [np.float64(0.0)]
locality_acc: [np.float64(1.0)]
[8/500] In Marshall Islands, the language spoken is a mixture of -> English
rewrite_acc: [np.float64(1.0)]
rephrase_acc: [np.float64(0.0)]
locality_acc: [np.float64(1.0)]
[9/500] Leslie Moonves is employed by -> CBS
rewrite_acc: [np.float64(1.0)]
rephrase_acc: [np.float64(0.0)]
locality_acc: [np.float64(1.0)]
[10/500] The original language of De finibus bonorum et malorum is the same as the -> Latin
rewrite_acc: [np.float64(1.0)]
rephrase_acc: [np.float64(0.0)]
locality_acc: [np.float64(1.0)]
... omitted 490 additional samples in console output

===== MEMIT Batch Editing Finished =====
Elapsed seconds: 2083.7560
Peak GPU memory (MB): 3603.19
Saved to: results\memit_results.json

(easyedit) E:\course\LLM\03>

```

ES/Efficacy: 100.00%

PS/Generalization: 39.90%

NS/Locality: 100.00%

## 5.3 结果分析

MEMIT 在 500 条批量编辑任务中实现了 100% 的编辑成功率，说明其批量知识写入能力是有效的。但与小规模编辑相比，本次实验中模型对改写问题的泛化能力明显下降，PS 仅为 39.9%。与此同时，局部性达到 100%，说明在该批量设置下，模型对无关知识保持得较好。整体来看，MEMIT 在大规模批量编辑中更强调局部性保持，但对同义改写的泛化能力仍然有限。

## 6. Task 4：综合评估

为了更全面地比较不同知识编辑方法，本实验使用编辑成功率（ES）、泛化性（PS）和局部性（NS）三个指标进行评估。

### 6.1 指标说明

ES (Efficacy)：模型在直接编辑 prompt 上是否能够输出目标答案。

PS (Generalization)：模型在同义改写 prompt 上是否仍能输出目标答案。

NS (Locality)：模型在编辑某条知识后，对无关事实的回答是否仍然保持正确。

## 6.2 综合结果表

### 10 条样本任务（基线+ROME）

| 方法       | ES/Efficacy | PS/Generalization | NS/Locality |
|----------|-------------|-------------------|-------------|
| Baseline | 0.00%       | –                 | –           |
| MEMIT    | 100.00%     | 76.57%            | 48.33%      |

### 500 条样本任务（MEMIT）

| 方法    | ES/Efficacy | PS/Generalization | NS/Locality |
|-------|-------------|-------------------|-------------|
| MEMIT | 100.00%     | 39.90%            | 100.00%     |

```
(easysedit) E:\course\LLM\03>python evaluate.py
===== Baseline Summary =====
Total samples: 10
Already matches target_new: 0/10 (0.0%)
Matches ground_truth: 6/10 (60.0%)

===== ROME Summary =====
Total samples: 10
ES / Efficacy: 100.0%
PS / Generalization: 76.57%
NS / Locality: 48.33%
Pre rewrite acc: 44.26%
Pre rephrase acc: 33.19%

===== Per-case ROME Metrics =====
ID      ES      PS      NS      Prompt
1  100.00%  85.71%  50.00%  The CEO of Starbucks is
2  100.00%  50.00%  50.00%  The CEO of Intel is
3  100.00%  50.00%  100.00% The monarch of the United Kingdom is
4  100.00%  100.00% 100.00% The CEO of Twitter is
5  100.00%  100.00% 66.67%  The head coach of FC Barcelona is
6  100.00%  100.00% 50.00%  The CEO of YouTube is
7  100.00%  100.00%  0.00%  The CEO of IBM is
8  100.00%  0.00%   0.00%  The capital of Turkey is
9  100.00%  80.00%  66.67%  The CEO of Google is
10 100.00%  100.00%  0.00%  The CEO of AMD is

===== MEMIT Summary =====
Total samples: 500
ES / Efficacy: 100.0%
PS / Generalization: 39.9%
NS / Locality: 100.0%
Pre rewrite acc: 46.3%
Pre rephrase acc: 0.1%
Dataset source: local cache: data\memit_500.json
Elapsed seconds: 2083.756
Peak GPU memory (MB): 3603.19

===== Per-case MEMIT Metrics =====
ID      ES      PS      NS      Prompt
1  100.00%  0.00%  100.00%  Vinson Massif is located in the continent of
2  100.00%  0.00%  100.00%  Beats Music is owned by
3  100.00%  0.00%  100.00%  Audible.com is owned by
4  100.00%  100.00% 100.00%  The Big Bang Theory premieres on
5  100.00%  100.00% 100.00%  MacApp, a product created by
6  100.00%  100.00% 100.00%  Giuseppe Angeli, who has a citizenship of
7  100.00%  0.00%  100.00%  Catalonia belongs to the continent of
8  100.00%  0.00%  100.00%  In Marshall Islands, the language spoken is a mixture of
9  100.00%  0.00%  100.00%  Leslie Moonves is employed by
10 100.00%  0.00%  100.00%  The original language of De finibus bonorum et malorum is the same as the
11 100.00%  100.00% 100.00%  Kirkpatrick Glacier belongs to the continent of
12 100.00%  100.00% 100.00%  The headquarter of Army of the Guardians of the Islamic Revolution is in
13 100.00%  100.00% 100.00%  Il Gazzettino was written in the early 1980s, when the
14 100.00%  100.00% 100.00%  Xamarin, from the
15 100.00%  100.00% 100.00%  Eavan Boland was born in
16 100.00%  0.00%  100.00%  Comme j'ai mal is written in
17 100.00%  100.00% 100.00%  The language used by Juan Bautista de Anza is a bit different from the language used by the
18 100.00%  0.00%  100.00%  Alfred Hitchcock Presents debuted on
19 100.00%  0.00%  100.00%  Tizen is developed by a consortium of companies including
20 100.00%  0.00%  100.00%  Honus Wagner professionally plays the sport of
... omitted 480 additional rows

Saved metrics to: results\metrics.json
(easysedit) E:\course\LLM\03>python evaluate.py
```

### 6.3 综合分析

从综合结果可以看出：1、ROME 和 MEMIT 都能够稳定完成目标知识写入，编辑成功率均达到 100%；2、ROME 在小规模单条编辑任务中泛化性较好；3、MEMIT 在大规模批量编辑任务中局部性表现更强，但对同义改写的泛化能力较弱；4、不同规模下，知识编辑方法在“编辑成功率、泛化能力和局部性”之间存在明显权衡。

## 7. 失败案例分析

在实验中，比较典型的失败样本是：The capital of Turkey is -> Ankara，在这一样本上，ROME 和 MEMIT 都表现为：ES=100%，PS=0%，NS=0%。

这说明模型虽然能够在原始 prompt 上输出目标答案 Ankara，但在改写问题上完全无法泛化，同时对无关知识也产生了较明显的影响。也就是说，这条知识虽然被“写入”到了模型中，但没有形成稳定的、可迁移的知识表示。除此之外，部分企业类事实（如 IBM、Google、AMD）的 locality 得分也偏低，说明知识编辑会对语义相近或相关联的知识产生一定干扰。

## 8. 实验总结

通过本次实验，我完成了从基线测试到单条知识编辑、批量知识编辑以及综合评估的完整流程，也对知识编辑方法有了更直观的理解。

### 实验结果表明：

- 1、原始模型不具备目标新知识；
- 2、ROME 和 MEMIT 都能有效修改模型中的目标事实；
- 3、ROME 在泛化能力上略有优势；
- 4、MEMIT 在局部性方面略有优势；
- 5、知识编辑虽然能够成功注入新知识，但仍可能对无关知识产生影响。

总体来看，本实验验证了知识编辑方法在小型语言模型上的可行性，也说明在实际应用中需要在编辑成功率、泛化性和局部性之间进行权衡。

## Bonus：跨语种泛化测试

### 9.1 实验目的

在完成基础知识编辑任务后，我进一步尝试了 Bonus 中的“跨语种泛化测试”。该任务的核心目标是：先使用英文事实对模型进行知识编辑，再使用中文问题测试模型能否正确输出对应的新知识。这样可以检验知识编辑后模型是否真正形成了跨语言层面的知识迁移能力，而不仅仅是在英文 prompt 上完成表面匹配。

### 9.2 实验设计

本实验额外构建了一个跨语种知识编辑数据集 data/crosslingual\_edits.json，共包含 12 条样本。每条样本同时包含：英文编辑 prompt、英文改写问题、中文改写问题、局部性测试问题、目标英文答案、目标中文答案。



例如，在英文中向模型注入：The capital of Australia is -> Canberra，然后再使用中文问题测试：澳大利亚的首都是哪里？

为了完成该部分实验，实现了以下三个脚本：bonus\_crosslingual\_rome.py、bonus\_crosslingual\_memit.py、bonus\_crosslingual\_evaluate.py。

其中：bonus\_crosslingual\_rome.py：使用 ROME 逐条编辑跨语种样本；

bonus\_crosslingual\_memit.py：使用 MEMIT 对跨语种样本进行批量编辑；

bonus\_crosslingual\_evaluate.py：统计英文编辑成功率、英文改写成功率、中文改写成功率、局部性，以及跨语种差距等指标。

此外，实验结果截图为：



```

(easyedit) E:\course\LLM\03>python bonus_crosslingual_evaluate.py

===== ROME Cross-lingual Summary =====
Total samples: 12
ES / English rewrite success: 100.0%
PS_en / English paraphrase success: 69.78%
PS_zh / Chinese paraphrase success: 8.33%
NS / Locality: 49.1%
Cross-lingual gap: 61.45%
Bilingual success rate: 8.33%

===== Per-case ROME Cross-lingual Metrics =====
ID      ES      EN      ZH      NS      BI      Prompt
1      100.00%    0.00%    0.00%    0.00%    0.00%    The capital of Turkey is
2      100.00%   100.00%  100.00%    0.00%   100.00%    The capital of Australia is
3      100.00%   50.00%    0.00%   100.00%    0.00%    The capital of Brazil is
4      100.00%   50.00%    0.00%   50.00%    0.00%    The president of France is
5      100.00%   80.00%    0.00%   25.00%    0.00%    The president of South Korea is
6      100.00%  100.00%    0.00%   87.50%    0.00%    The president of Mexico is
7      100.00%   75.00%    0.00%   50.00%    0.00%    The prime minister of the United Kingdom is
8      100.00%  100.00%    0.00%   60.00%    0.00%    The prime minister of India is
9      100.00%   50.00%    0.00%   50.00%    0.00%    The monarch of the United Kingdom is
10     100.00%   80.00%    0.00%   66.67%    0.00%    The CEO of Google is
11     100.00%   66.67%    0.00%   50.00%    0.00%    The CEO of YouTube is
12     100.00%   85.71%    0.00%   50.00%    0.00%    The CEO of Starbucks is
MEMIT cross-lingual result not found, skip: results\memit_crosslingual_results.json

Saved cross-lingual metrics to: results\crosslingual_metrics.json

(easyedit) E:\course\LLM\03>

```

[illegible]

```

(easyedit) E:\course\LLM\03>python bonus_crosslingual_evaluate.py

===== ROME Cross-lingual Summary =====
Total samples: 12
ES / English rewrite success: 100.0%
PS_en / English paraphrase success: 69.78%
PS_zh / Chinese paraphrase success: 8.33%
NS / Locality: 49.1%
Cross-lingual gap: 61.45%
Bilingual success rate: 8.33%

===== Per-case ROME Cross-lingual Metrics =====
ID      ES      EN      ZH      NS      BI      Prompt
1      100.00%    0.00%    0.00%    0.00%    0.00%    The capital of Turkey is
2      100.00%   100.00%   100.00%    0.00%   100.00%    The capital of Australia is
3      100.00%    50.00%    0.00%   100.00%    0.00%    The capital of Brazil is
4      100.00%    50.00%    0.00%    50.00%    0.00%    The president of France is
5      100.00%    80.00%    0.00%    25.00%    0.00%    The president of South Korea is
6      100.00%   100.00%    0.00%   87.50%    0.00%    The president of Mexico is
7      100.00%    75.00%    0.00%    50.00%    0.00%    The prime minister of the United Kingdom is
8      100.00%   100.00%    0.00%    60.00%    0.00%    The prime minister of India is
9      100.00%    50.00%    0.00%    50.00%    0.00%    The monarch of the United Kingdom is
10     100.00%    80.00%    0.00%   66.67%    0.00%    The CEO of Google is
11     100.00%   66.67%    0.00%    50.00%    0.00%    The CEO of YouTube is
12     100.00%   85.71%    0.00%    50.00%    0.00%    The CEO of Starbucks is

===== MEMIT Cross-lingual Summary =====
Total samples: 12
ES / English rewrite success: 100.0%
PS_en / English paraphrase success: 69.78%
PS_zh / Chinese paraphrase success: 8.33%
NS / Locality: 53.26%
Cross-lingual gap: 61.45%
Bilingual success rate: 8.33%
Elapsed seconds: 72480.2427
Peak GPU memory (MB): 7878.27

===== Per-case MEMIT Cross-lingual Metrics =====
ID      ES      EN      ZH      NS      BI      Prompt
1      100.00%    0.00%    0.00%    0.00%    0.00%    The capital of Turkey is
2      100.00%    0.00%   100.00%    0.00%   100.00%    The capital of Australia is
3      100.00%   100.00%    0.00%    50.00%    0.00%    The capital of Brazil is
4      100.00%   100.00%    0.00%    50.00%    0.00%    The president of France is
5      100.00%    80.00%    0.00%    50.00%    0.00%    The president of South Korea is
6      100.00%   100.00%    0.00%   87.50%    0.00%    The president of Mexico is
7      100.00%    75.00%    0.00%    50.00%    0.00%    The prime minister of the United Kingdom is
8      100.00%   100.00%    0.00%    60.00%    0.00%    The prime minister of India is
9      100.00%    50.00%    0.00%    50.00%    0.00%    The monarch of the United Kingdom is
10     100.00%    80.00%    0.00%   66.67%    0.00%    The CEO of Google is
11     100.00%   66.67%    0.00%   75.00%    0.00%    The CEO of YouTube is
12     100.00%   85.71%    0.00%    50.00%    0.00%    The CEO of Starbucks is

```

### 9.3 评估指标

在基础 ES/PS/NS 的基础上，跨语种实验进一步区分了英文和中文场景，主要使用以下指标：ES：英文直接编辑 prompt 的成功率；PS\_en：英文改写问题的成功率、PS\_zh：中文改写问题的成功率；NS：局部性保持情况、Cross-lingual gap：英文改写成功率与中文改写成功率之间的差距；Bilingual success rate：同一样本在英文和中文改写上都回答正确的比例。

### 9.4 实验结果

根据 results/crosslingual\_metrics.json 的统计结果：

| 方法    | ES      | PS_en  | PS_zh | NS     | Cross-lingual gap | Bilingual success rate |
|-------|---------|--------|-------|--------|-------------------|------------------------|
| ROME  | 100.00% | 69.78% | 8.33% | 49.10% | 61.45%            | 8.33%                  |
| MEMIT | 100.00% | 69.78% | 8.33% | 53.26% | 61.45%            | 8.33%                  |

## 9.5 结果分析

实验结果表明，ROME 和 MEMIT 在跨语种场景下都能够保持 100% 的英文直接编辑成功率，说明英文事实注入本身是有效的。同时，两种方法在英文改写问题上也有一定表现，PS\_en 均达到 69.78%，说明编辑后的知识在英文内部具有一定泛化能力。

但是，当问题切换到中文后，结果明显下降：PS\_zh=8.33%、Bilingual success rate=8.33%、Cross-lingual gap=61.45%。这说明模型虽然在英文中完成了知识编辑，但这些新知识并没有有效迁移到中文提问场景中。也就是说，模型学到的更多是英文 prompt 下的局部映射关系，而不是可以跨语言共享的稳定知识表示。

从局部性指标来看，MEMIT 略优于 ROME：ROME：49.10%、MEMIT：53.26%。这一现象与主实验中的结果相似，即 MEMIT 在保持无关知识稳定性方面略好一些，但这种优势并没有转化为更好的跨语种泛化能力。

## 9.6 典型现象与失败案例

从逐样本结果中可以观察到，许多样本在英文上表现正常，但中文回答仍然回到旧知识或错误知识。例如：The capital of Turkey is -> Ankara、英文改写问题可以回答 Ankara、中文问题“土耳其的首都是哪里？”仍然输出“伊斯坦布尔”。再例如：The president of France is -> Emmanuel Macron、中文问题中模型虽然输出“马克龙”，但并未严格对应目标中文全名“埃马纽埃尔·马克龙”。这些现象说明，模型在知识编辑后并没有稳定建立中英文共享的事实表示，而更像是在英文表达范围内局部更新了参数。

## 9.7 小结

通过跨语种泛化测试可以看出，当前知识编辑方法虽然能够较好地完成英文场景下的知识写入，但其跨语言迁移能力仍然较弱。对于本实验所使用的 Qwen2.5-0.5B-Instruct 模型，无论是 ROME 还是 MEMIT，在中文改写问题上的表现都明显低于英文改写问题。

因此，可以得出如下结论：1、英文知识编辑成功并不代表中文场景下也能同步生效；2、当前知识编辑方法更像是在特定语言表达空间内完成局部修改；3、如果希望实现更强的跨语种知识迁移，还需要设计更适合多语言共享表示的编辑方法，或者在编辑时显式加入多语种监督。